# Huawei Access Node ELEC-ACC-57 QoS Congestion Event SOP

**Node:** ELEC-ACC-57 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_4496

## Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution steps for addressing QoS congestion events on the Huawei Access Node ELEC-ACC-57 at the Electronic City Hub site. The failure is attributed to microbursts, interface tail drops, and QoS queue starvation, which can occur due to the Node's role as an Access device and the Huawei CLI dialect used. The Huawei Access Node is responsible for managing and controlling the network traffic flow, ensuring that packets are transmitted and received efficiently. However, in the event of a QoS congestion event, the Node's ability to manage traffic becomes compromised, leading to microbursts, interface tail drops, and QoS queue starvation. To mitigate this issue, it is essential to diagnose the root cause of the congestion event and resolve it promptly to prevent further network disruptions.

## Pre-requisite Checks

- check: Verify that the site is physically accessible and that all necessary permissions are obtained before proceeding with the investigation.

- check: Confirm that the redundant link is operational and can handle the network traffic in case of a failure.

- check: Ensure that the Huawei Access Node ELEC-ACC-57 is properly powered and all interfaces are online.

## Potential Root Causes

| Cause | Probability |
|---|---|
| Inadequate QoS configuration | High |
| Excessive microburst traffic | Medium |
| Interface congestion due to high traffic volume | Low |

## Diagnostic Commands & Expected Outputs

```
Command: display qos queue statistics, Expected Output: Queue statistics for QoS queue 1:, Description: This command is used to gather information about the QoS queue statistics, including the queue size, packet drop count, and packet transmit count.

Command: display interface traffic statistics, Expected Output: Interface traffic statistics for GigabitEthernet 0/1/0:, Description: This command is used to gather information about the interface traffic statistics, including the inbound and outbound packet counts, byte counts, and error counts.
```

```
Command: display qos policy statistics, Expected Output: QoS policy statistics for policy 10:, Description: This command is used to gather information about the QoS policy statistics, including the policy name, queue number, and packet drop count.
```

## Resolution Steps

Step 1: Step: Adjust QoS configuration to prioritize critical traffic, Command: qos policy 10 priority-class 1 bandwidth 1000000, Description: This command adjusts the QoS policy to prioritize critical traffic by increasing the bandwidth allocation for priority-class 1.

Step 2: Step: Configure microburst suppression, Command: interface GigabitEthernet 0/1/0 microburst suppression enable, Description: This command enables microburst suppression on the specified interface to prevent excessive packet bursts.

Step 3: Step: Adjust interface congestion threshold, Command: interface GigabitEthernet 0/1/0 congestion threshold 1000000, Description: This command adjusts the interface congestion threshold to prevent excessive packet drops.

## Post-Check Verification

```
Command: display qos queue statistics, Expected Output: Queue statistics for QoS queue 1: no packet drops, Description: This command verifies that the QoS queue statistics show no packet drops.

Command: display interface traffic statistics, Expected Output: Interface traffic statistics for GigabitEthernet 0/1/0: no errors, Description: This command verifies that the interface traffic statistics show no errors.

Command: display qos policy statistics, Expected Output: QoS policy statistics for policy 10: no packet drops, Description: This command verifies that the QoS policy statistics show no packet drops.
```

## Roll-back Procedures

- Step: Restore original QoS configuration, Command: undo qos policy 10 priority-class 1 bandwidth 1000000, Description: This command restores the original QoS configuration by rolling back the changes made to the priority-class 1 bandwidth allocation.

- Step: Disable microburst suppression, Command: interface GigabitEthernet 0/1/0 microburst suppression disable, Description: This command disables microburst suppression on the specified interface to restore the original settings.

- Step: Restore original interface congestion threshold, Command: undo interface GigabitEthernet 0/1/0 congestion threshold 1000000, Description: This command restores the original interface congestion threshold to prevent excessive packet drops.